

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third & Fourth Semester of B. Tech. Examination (EE)

May 2014

EE-201 CONTROL ENGINEERING

Date: 02.05.2014, Friday

Time: 10.00 a.m. To 1.00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Choose the right answers from the given choices.

[03]

(i) Which system are less sensitive to noise present in the system?

- | | |
|----------------------------|--------------------------|
| (a) Linear System | (b) Non- Linear System |
| (c) Continuous Time system | (d) Discrete Time system |

(ii) The characteristics equation of an armature controlled DC servomotor is of

- | | |
|--------------------------|---------------------------|
| (a) First Order Equation | (b) Second Order Equation |
| (c) Third Order Equation | (d) None of Above |

(iii) The Output signal of any controller is

- | | |
|----------------------|---------------------|
| (a) Error signal | (b) Feedback Signal |
| (c) Actuating Signal | (d) Output Signal |

(b) Define Type and Order of the system.

[01]

(c) A system is described by

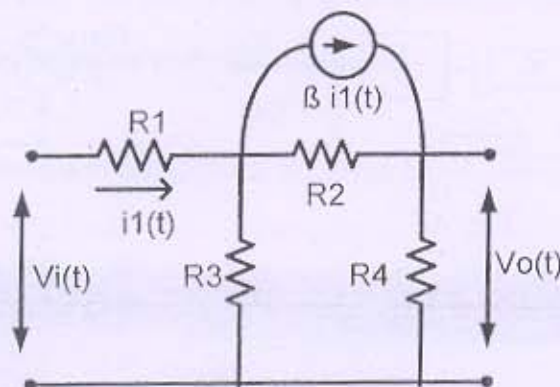
[03]

$$\dot{x} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u(t)$$

$$y = [1 \quad 0 \quad 2] x(t) \quad \text{Check Controllability or Observability.}$$

Q - 2 (a) Find the Transfer function of the system using network equations.

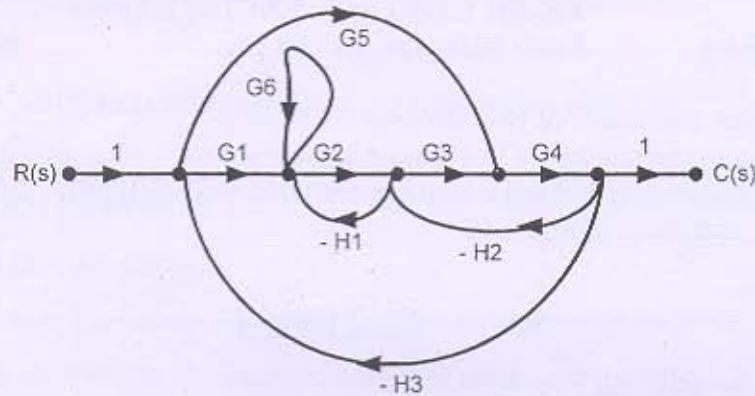
[07]



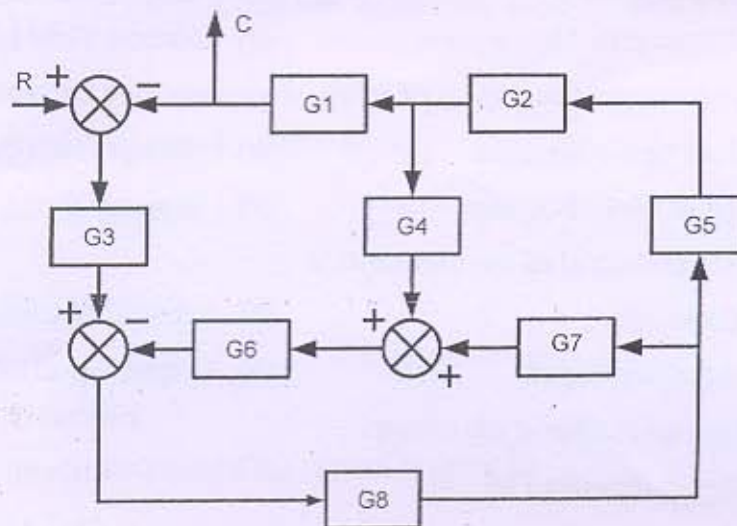
- (b) What do you mean by Servomechanism? Explain the Servomechanism used as position control system with neat diagram. [07]

OR

- Q - 2 (a) Find Transfer function of the signal flow graph using Mason's Gain Formula. [07]



- (b) Find out Transfer Function using Block Diagram Reduction method. [07]



- Q - 3 (a) Draw the typical block diagram of closed loop control system, explain each block in detail. [07]
 (b) For a control system shown in Fig. A, find values of K_1 , K_2 so that $M_p = 25\%$ and $T_p = 4$ sec. Assume unit step unit. [03]

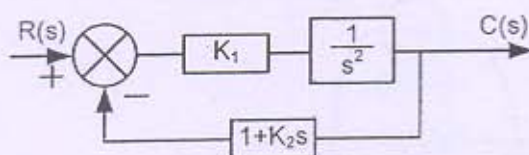


Fig. A

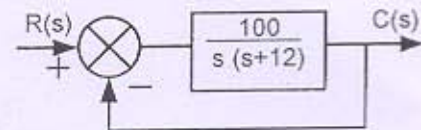


Fig. B

- (c) Explain in detail about standard inputs used in control engineering. Also tabulate their values in Time domain and in Laplace domain [04]

OR

- Q - 3 (a) Draw a typical response $c(t)$ with respect to time of 2nd Order system and define the following terms with their mathematical equations. [07]
- Delay Time
 - Rise Time
 - Settling Time
 - Peak overshoot
 - Peak Time
 - Steady State Error
- (b) For the system shown in Fig. B, determine % Mp and Ts when it is excited by unit step input. If for the same system, PD controller having a constant $K_p=1, K_d = 1/30$ is used in forward path, determine new values of damping ratio, Mp and Ts. [07]

SECTION – II

- Q - 4 (a) Choose the right answers from the given choices. [03]
- (i) Choose a good closed loop control system depending upon the value of ξ
- (a) $\xi = 0$ (b) $\xi = 1$
 (c) $0 < \xi < 1$ (d) $1 < \xi < \infty$
- (ii) The starting point of a root locus is
- (a) Open loop pole (b) Closed loop pole
 (c) Open loop zero (d) Closed loop zero
- (b) Define Dominant Poles & Insignificant Poles. [01]
- (c) Why do we require compensation in control engineering? Explain any compensation technique. [03]
- Q - 5 (a) Sketch the root locus of the system with [07]

$$G(s)H(s) = \frac{K}{s(s+1)(s+3)}$$

Where K is varying from zero to infinity.

- (b) The characteristics equation of the system is given by [07]
- $$s^6 + 2s^5 + 7s^4 + 10s^3 + 14s^2 + 8s + 8 = 0.$$
- Find the stability of the system using R-H Criteria.

OR

- Q - 5 (a) The characteristics equations of the systems are given by [07]
- (i) $s^3 + 5s^2 + 10s + 3 = 0$
 (ii) $s^3 + 2s^2 + 3s + 10 = 0.$
- Find the stability of the systems using R-H Criteria.

- (b) Sketch the root locus of the system with

[07]

$G(s)H(s) = \frac{K}{s(s+2)(s+4)}$ where K is varied from 0 to infinity. And obtain the value of K for which the system is unstable.

- Q - 6 (a) Draw the Bode plot for the unity feedback control system given by

[07]

$$G(s) = \frac{36(1+0.2s)}{s^2(1+0.05s)(1+0.01s)}$$

From the Bode plot determine.

(a) Gain cross over frequency

(b) Phase cross over frequency

Also Comment on stability

- (b) For a feedback control system ,

[07]

$$G(s)H(s) = \frac{60}{(s+1)(s+2)(s+5)}$$
 Comment on stability using Nyquist Plot.

OR

- Q - 6 (a) Draw the Bode plot for the transfer function.

[07]

$$G(s) = \frac{40}{s(s+2)(s+5)} \text{ \& } H(s) = 1$$

From the Bode plot determine.

(a) Gain Margin

(b) Phase Margin

Also Comment on stability

- (b) The Open loop transfer function of the system is given by,

[07]

$$G(s)H(s) = \frac{10(1+T_1s)}{s^2(1+T_2s)}$$
 Sketch the Nyquist plot for

(i) $T_1 > T_2$ (ii) $T_1 < T_2$ (iii) (i) $T_1 = T_2$

Comment on stability.
